

Lecture 3: Elasticities & Taxes — the Supply & Demand model II (Ch. 2 recap · §6.4)

Applied Microeconomics — Master Notes · by Jakob V. Stangel

Required reading: Friberg, §6.4 Elasticities (own-price, cross-price, income; supply elasticity) — building on the Chapter 2 supply-and-demand model (see Lecture 2). The lecture deck also teaches **taxes on goods & services** — that part draws on the textbook's §7.5 (unit tax, tax incidence), covered here because the deck foregrounds it.

Theme of the lecture (Butera — from the deck, titled "*Supply and Demand model*"). Chapter 2 taught us to **move the curves around**; this lecture adds **precision**. The *"shape"* of a demand or supply curve — its **elasticity** — decides how much a shock (or a tax) shows up as a change in price versus a change in quantity. The deck's three goals: (1) understand what demand & supply elasticities mean, (2) derive them from demand/supply curves, and (3) use them to analyse the **effects of taxes**.

★ **The heart of the lecture:** elasticity = % responsiveness — "by what % does quantity change when price rises 1%." It's a **unit-free** number that summarises a curve's **shape**, so it's comparable across markets and time. The **shape decides how a shock splits into price vs quantity**: with **inelastic** demand a shock/tax hits **price**; with **elastic** demand it hits **quantity**. And a **tax** drives a **wedge** between the price buyers pay and sellers receive — **who bears it depends on the relative elasticities**, and (in perfect competition) **it doesn't matter who is legally taxed**.

The red thread. For any market ask: *how responsive is quantity to price (elastic or inelastic)?* — because that single question answers **how a supply shock moves the market**, **how a firm should price**, and **who really pays a tax**. Elasticity is the "shape" number that ties all three together.

Key concepts / "modes" to use

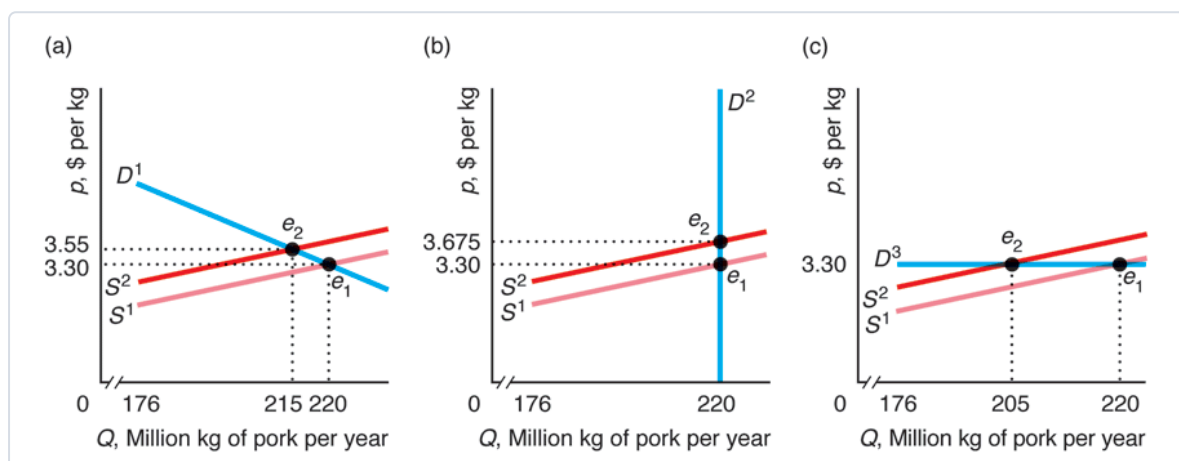
- **Elasticity** — the % change in one variable ÷ % change in another (a unit-free number).
- **Price elasticity of demand (E_D)** — % change in quantity demanded ÷ % change in own price (negative).
- **Elastic / inelastic / unit-elastic** — $|E| > 1$ / $|E| < 1$ / $|E| = 1$.
- **Point-elasticity trick** — $E_D = (dQ/dp) \cdot (p/Q)$; on a linear curve elasticity **changes along the curve** (constant-elastic curves don't).
- **Cross-price elasticity** — substitutes (+) vs complements (−).
- **Income elasticity** — normal (+) vs inferior (−); **Engel curves / Engel's law**.
- **Supply elasticity** — % change in quantity supplied ÷ % change in price (positive).
- **Tax wedge & tax incidence** — $p_b = p_s + t$; who bears the burden depends on **relative elasticities**; legal $\neq$ economic incidence; unit vs ad-valorem tax; **deadweight loss**.

Section-by-section main points

Why do we care about elasticity? (deck)

Core: the **shape** of demand/supply curves determines **how much a shift in one moves the equilibrium** — and "shape" is summarised by **elasticity**. A steep (inelastic) curve and a flat (elastic) curve respond to the *same* shock very differently.

- **The motivating questions (deck):** Do "sin taxes" (cigarettes, sugar) hurt the poor? Why do firms **raise prices** even when it looks crazy (Netflix, Amazon Prime)? To what extent are **taxes on producers "passed on"** to consumers? Every answer runs through elasticity.
- **The visual intuition:** how far a **supply shift** moves price vs quantity depends entirely on the **slope/elasticity of demand** — see the figure below (perfectly inelastic demand → only price moves; perfectly elastic demand → only quantity moves).



How to read it (why elasticity matters). All three panels apply the **same supply shift** $S^1 \rightarrow S^2$. The **only** thing that differs is the **demand curve's shape**:

- **(a) Ordinary downward-sloping demand:** the shock splits into **both** a higher price *and* a lower quantity.
- **(b) Perfectly inelastic demand (vertical):** quantity can't respond, so the shock lands **entirely on price** (3.30 → 3.675). *This is the "necessity with no substitutes" case.*
- **(c) Perfectly elastic demand (horizontal):** price can't move, so the shock lands **entirely on quantity** (220 → 205).
- **Take-away:** the **more inelastic** the demand, the more a shock (or tax) shows up as **price**; the **more elastic**, the more it shows up as **quantity**. *This one picture is the whole reason we measure elasticity.*

★ Price elasticity of demand — definition

Core: the **price elasticity of demand** measures how sensitive quantity demanded is to the good's own price — the **% change in quantity ÷ % change in price**. It is a **pure (unit-free) number** and is **negative** (demand slopes down: price up → quantity down).

PRICE ELASTICITY OF DEMAND

$$E_D = (\Delta Q / Q \times 100) \div (\Delta P / P \times 100) = \% \Delta Q \div \% \Delta P$$

Reads: how responsive quantity demanded is to a price change — "a 1% rise in price p leads to an E_D % change in quantity q ." — $\Delta Q/Q \times 100$ = the percentage change in quantity demanded · $\Delta p/p \times 100$ = the percentage change in price · E_D = the ratio (a unit-free number, negative for a normal good). The $\times 100$ s cancel, so it's just % ΔQ over % Δp .

- **Why unit-free matters (the point of the whole concept):** raw slopes aren't comparable — "1,000 tons less rice" means something different in **Vietnam** vs **Bhutan**, and depends on the currency. A percentage measure is **comparable across markets, currencies and time**.
- **Why negative:** demand curves slope **down**, so ΔQ and Δp have opposite signs → the ratio is negative. (When people say a "higher" elasticity they usually mean a bigger **absolute value** $|E_D|$.)
- **Worked example:** a café raises its coffee price **10%** and sells **8% fewer** cups → $E_D = -8\% \div +10\% = -0.8$. Since $|-0.8| < 1$, demand is **inelastic** — revenue actually **rises** (price up 10%, quantity down only 8%). (Full step-by-step computations are in the next section.)

Computing elasticity — and the point-elasticity trick

Core: plug the two percentage changes into the formula. **On a linear demand curve the elasticity is not constant** — it depends on *where* you evaluate it, via $E_D = (dQ/dp) \cdot (p/Q)$.

- **Simple example (textbook Table 6.2):** quantity falls **6** → **4** while price rises **1** → **2**. Then

WORKED — SIMPLE EXAMPLE

$$E_D = ((4-6)/6) \div ((2-1)/1) = (-2/6) \div (1) = -1/3$$

Reads: a demand that is **inelastic** — $(4-6)/6 = -0.33$ = the % fall in quantity (−33%) · $(2-1)/1 = 1$ = the % rise in price (+100%) · $-1/3$ = quantity falls **only 1/3** as fast as price rises → **inelastic** demand.

- **A policy-style example (soda, Allcott et al. 2019):** own-price elasticity of sugary soda ≈ -1.37 . If price rises 5%, quantity changes by $E_D \times 5\% = -1.37 \times 5 = -6.85\%$ (a ~7% fall). To cut demand 15%, price must rise $-15 \div -1.37 \approx 11\%$. → *this is exactly the sum a policymaker designing a soda tax needs.*
- **The point-elasticity trick (linear demand):** rewrite the formula as

POINT ELASTICITY

$$E_D = (\Delta Q/\Delta p) \cdot (p/Q) = (dQ/dp) \cdot (p/Q)$$

Reads: elasticity = **slope of the demand function** × the **price-to-quantity ratio** at your chosen point — dQ/dp = the **slope** (how many units Q changes per €1 of price; constant for a straight line) · p/Q = the **price-to-quantity ratio** where you evaluate it (this is what varies along the curve).

- **Worked ($Q = 10 - 2p$):** here $dQ/dp = -2$ (always). But the **elasticity varies**:
 - at $p = 2.5$ (so $Q = 5$): $E_D = -2 \times 2.5/5 = -1$ (**unit elastic**).
 - at $p = 4$ (so $Q = 2$): $E_D = -2 \times (p/Q) = -2 \times 4/2 = -4$ → **elastic**. (Δ The textbook misprints this as $-2 \times 2/2 = -2$ — it accidentally puts $Q=2$ in the numerator instead of $p=4$; the correct p/Q is $4/2 = 2$, so $E_D = -4$.)

- **The pattern: demand is more elastic at higher prices** — at $p=2.5$ it's unit-elastic (-1), at $p=4$ it's elastic (-4). A fixed unit change is a bigger *percentage* change when Q is small and p is large.
- **Constant-elasticity demand (contrast):** the form $Q = a \cdot p^\epsilon$ has the **same** elasticity ϵ at *every* price (used in empirical estimation via $\ln Q = \ln a + \epsilon \cdot \ln p + \text{error}$). So: **linear** → **elasticity changes along the curve**; **constant-elastic** → **it doesn't**.
- **Worked example:** with $Q = 100 \cdot p^{-1.2}$, the elasticity is -1.2 whether you evaluate at $p = €1$ ($Q = 100$), $p = €5$ ($Q \approx 15$), or $p = €50$ — it never changes. (Contrast the linear $Q = 10 - 2p$ above, where it swung from -1 to -4 .)

Elastic, inelastic, unit-elastic — and why it matters

Core: we name the ranges of elasticity, and each has a sharp economic meaning for how supply shocks split into price vs quantity.

Name	Level (of E_D)	Meaning	Curve looks...
Inelastic	$-1 < E_D < 0$ ($ E < 1$)	$\% \Delta Q$ smaller than $\% \Delta p$	relatively steep
Unit elastic	$E_D = -1$	$\% \Delta Q$ equals $\% \Delta p$	—
Elastic	$E_D < -1$ ($ E > 1$)	$\% \Delta Q$ bigger than $\% \Delta p$	relatively flat

- **Why it matters (the punchline):** with **inelastic** demand, **supply shocks translate into big price changes** (little quantity response) — critical for e.g. **cocoa** growers in Ghana/Côte d'Ivoire (world cocoa elasticity ≈ -0.19 to -0.96 ; policy point estimate -0.34). With **elastic** demand, shocks translate into **quantity** changes.
- **What makes demand elastic — substitutes.** The single biggest driver is **availability of close substitutes**: pharmaceuticals with few substitutes are **inelastic** (people pay whatever for a needed medicine); goods with many substitutes are **elastic**.
- **Aggregation level matters:** a *specific brand* is more **elastic** than the *category* (more substitutes for one brand). US beer: **product-level** median elasticity ≈ -4.74 , but beer **overall** ≈ -0.60 .

Cross-price elasticity — substitutes vs complements

Core: the **cross-price elasticity** measures how demand for good **A** responds to a change in the price of a *different* good **B**. Its **sign** tells you the *relationship* between the two goods; its **size** tells you *how strong* that relationship is.

CROSS-PRICE ELASTICITY

$$E_{cross} = \% \Delta Q_A \div \% \Delta p_B \rightarrow > 0 = \text{substitutes} \cdot < 0 = \text{complements} \cdot \approx 0 = \text{unrelated}$$

Reads: how the quantity of **A** responds to a price change in **B** — $\% \Delta Q_A$ = percentage change in demand for good A · $\% \Delta p_B$ = percentage change in the price of good B · E_{cross} = the ratio (can be + or −, unlike own-price which is always −).

- **The sign = the relationship:**
 - **Positive → substitutes.** B gets dearer → buyers switch to A → A's demand **rises** (butter ↑ → margarine demand ↑; Heineken ↑ → Carlsberg ↑).
 - **Negative → complements.** B gets dearer → buyers use less B *and* less of its partner A → A's demand **falls** (petrol ↑ → big-car demand ↓; smartphones ↑ → phone-case demand ↓).
 - **≈ 0 → unrelated goods** (the price of bread barely touches demand for haircuts).
- **The size = the strength:** the larger the **absolute value** $|E_{\text{cross}}|$, the **closer** the substitutes (or the stronger the complementarity). Two brands of the *same* soda have a **big** positive cross-price elasticity; soda and bottled water a **small** one; soda and umbrellas ≈ 0.
- **Worked example: Danish spirits get 10% dearer** → Swedish demand for spirits changes by $E_{\text{cross}} \times 10\% = +0.3 \times 10 = +3\%$ (Swedes buy **more at home** instead of driving to Denmark) → **positive → substitutes** (Asplund 2007). *Flip it: if smartphones get 10% dearer and phone-case demand falls 6%, $E_{\text{cross}} = -6 \div 10 = -0.6$ → negative → complements.*
- ★ **Its killer application — defining markets & judging mergers.** This is *the* reason cross-price elasticity matters in practice. Competition authorities use it to decide **whether two products are in the same market**: a **high positive** cross-price elasticity means A and B are **close substitutes**, so a firm selling both has real market power (a merger between them removes a competitor). This is the logic of the **SSNIP** / "**hypothetical monopolist**" test — *would a Small but Significant Non-transitory Increase in Price (say 5%) be profitable, or would enough buyers flee to substitutes to make it a loss?* If they flee, the substitute is in the same market.
- **Asymmetry (watch out):** cross-price elasticity is **not symmetric** — A's response to B's price ≠ B's response to A's price. A cheap **complement** (phone case) reacts strongly to the price of the expensive main good (phone), but the phone barely reacts to the case's price.
- **Real numbers (Asplund et al. 2007):** Swedish **spirits +0.3, beer +0.5, wine +0.2** with respect to their **Danish** prices — all **positive**, i.e. all **substitutes** for cross-border shopping (beer the closest). (*Deck aside: cigarettes & alcohol ≈ -1 — strong complements, Krauss et al. 2014.*)

Income elasticity — normal vs inferior goods, Engel curves

Core: the **income elasticity** measures how demand responds to **income** — its **sign** distinguishes **normal** from **inferior** goods.

INCOME ELASTICITY

$$E_{\text{income}} = \% \Delta Q \div \% \Delta I \rightarrow > 0 = \text{normal good} \cdot < 0 = \text{inferior good}$$

Reads: how quantity demanded responds to an income change — $\% \Delta Q$ = percentage change in quantity · $\% \Delta I$ = percentage change in income · **sign: positive → normal** (demand rises with income), **negative → inferior** (demand falls as income rises — e.g. instant noodles, store-brand goods).

- **Worked example:** a household's **income rises 10%** and its spirits purchases rise 14% → $E_{\text{income}} = +14 \div +10 = +1.4$ → **normal and income-elastic** (a "luxury," > 1) (Asplund 2007). *Flip it: income +10% but instant-noodle demand -5% → $E_{\text{income}} = -5 \div 10 = -0.5$ → inferior.*

- **Nuance:** a good can be **normal at low incomes and inferior at higher ones** (hostels for a student: more nights as income rises, then switch to hotels). Tastes vary by person.
- **Uses:** long-term **projections** (healthcare, appliances in low-income countries) and predicting the **business cycle** (pharma is income-insensitive → "defensive stock").
- **Engel curves / Engel's law (Ernst Engel, 1800s Prussian statistician):** plot demand (vertical) against income (horizontal) — **positive slope for normal goods, negative for inferior**. **Engel's law:** the **share of income spent on food falls as income rises** → the income elasticity of food is **less than 1**.

Supply elasticity — same principle

Core: the **price elasticity of supply** measures how sensitive quantity *supplied* is to price — same formula, but **positive** (supply slopes up).

PRICE ELASTICITY OF SUPPLY

$$E_S = \% \Delta Q_S \div \% \Delta p \quad (\text{positive})$$

Reads: how responsive quantity supplied is to a price change — $\% \Delta Q_S$ = percentage change in quantity supplied · $\% \Delta p$ = percentage change in price · E_S = the ratio (**positive**: higher price → more supplied). A **steep** supply curve is **inelastic** (hard to expand output quickly); a **flat** one is **elastic**.

- **Worked example:** the market price of new flats rises **20%**, but the quantity of new flats supplied rises only **4%** this year → $E_S = +4 \div +20 = +0.2$ → **inelastic** supply (you can't build quickly). Give it several years and builders respond more → supply becomes **more elastic**.

Elasticity over time — short run vs long run

Core: demand (and supply) are usually **more elastic in the long run** than the short run, because people have **more time to adjust / find substitutes**.

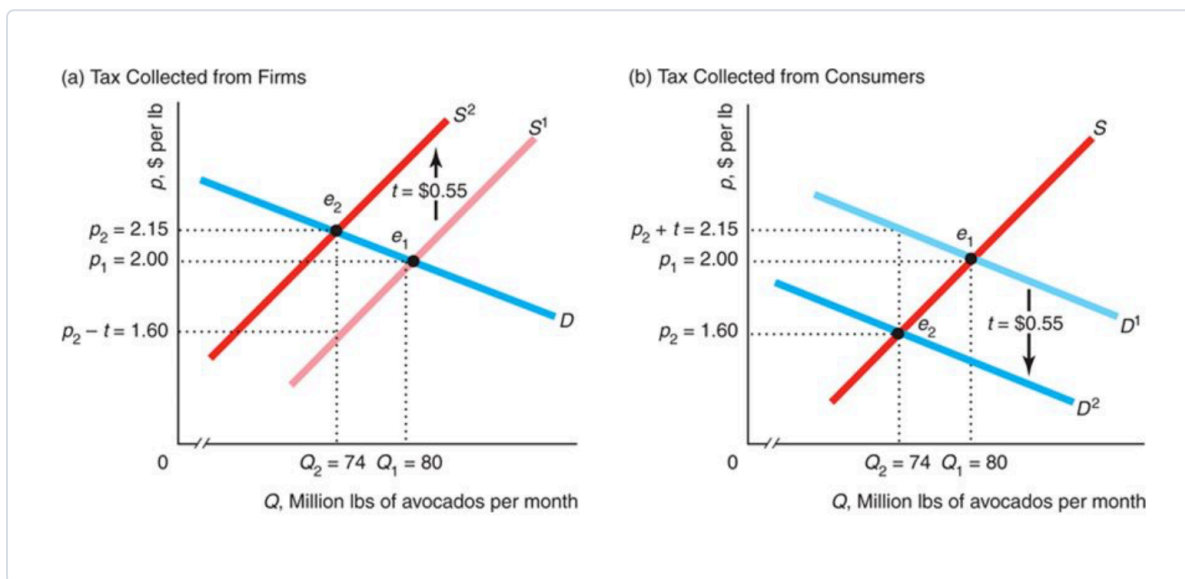
- **Electricity (Labandeira et al. 2017):** ≈ -0.24 short-run vs -0.6 long-run — over time consumers buy more efficient equipment, change habits.
- **The deck's cases:** a drop in **gas prices** does little in the short run but more in the long run; likewise **computers**. *Exam framing:* "in the short run demand is inelastic (people are locked in); given time they substitute, so long-run demand is more elastic."

★ Taxes on goods & services — the wedge, incidence & equivalence (deck; textbook §7.5)

Core: a tax **drives a wedge** between the price **buyers pay** (p_b) and the price **sellers receive** (p_s): $p_b = p_s + t$. It raises the buyer price, lowers the seller price, cuts the quantity traded, and creates a **deadweight loss**. The split of the burden — **tax incidence** — depends on the **relative elasticities**, and (under perfect competition) **not** on who is legally taxed.

- **Two types of tax (deck):**
 - **Unit (specific) tax** — a fixed amount t per unit (per litre of petrol, per cigarette).

- **Ad valorem tax** ("sales tax"/VAT) — a **fraction α of the price/spend** (government keeps a share of each euro spent).
- **The wedge (worked unit-tax example, $t = €3$)**: with $Q_D = 23 - p_b$, $Q_S = -1 + p_s/2$, and $p_b = p_s + 3$, solving gives $p_s = 14$, $p_b = 17$, $Q = 6$. Buyers pay **€3 more** than sellers receive; quantity falls from the no-tax equilibrium.
- **Tax revenue** = $t \times Q = 3 \times 6 = €18$. The rest of the surplus lost is **deadweight loss** (here $3 \times (7-6)/2 = €1.5$) — trades that *would* have happened but now don't.
- **Legal incidence $\neq$ economic incidence (the key result)**. It **doesn't matter** whether the tax is collected from **firms** (supply shifts up by t) or **consumers** (demand shifts down by t) — the quantity, the prices actually paid/received, and the DWL are **identical**. *All that matters is that the tax drives a wedge.* (Perfect-competition assumptions make this exact: no transaction costs, perfect information.)



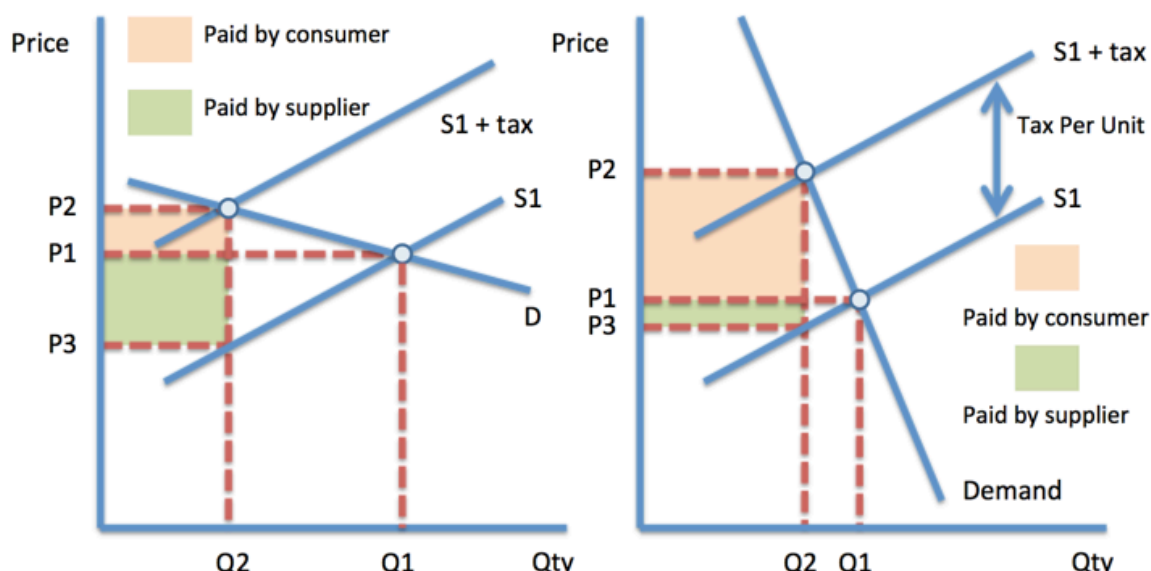
How to read it (tax equivalence). Both panels impose the **same \$0.55 tax** on avocados. (a) collects it from **firms** → the supply curve shifts **up** by t ; (b) collects it from **consumers** → the demand curve shifts **down** by t . In **both** cases the result is identical: **Q falls to 74, buyers pay 2.15, sellers receive 1.60**, and the wedge between them is exactly **\$0.55**. → *The economics doesn't care who writes the cheque.*

- **What *does* decide the split: relative elasticities.** The **more inelastic** side of the market bears **more** of the tax (it can't escape by changing quantity). The **more price-sensitive (elastic)** side is affected **less** on price, **more** on quantity.

Indirect Taxes with Different Coefficient of PED

If the co-efficient of price elasticity of demand >1 , then most of the burden of an indirect tax will be absorbed by the supplier

If the co-efficient of price elasticity of demand <1 , most of an indirect tax can be passed on to the final consumer



How to read it (incidence & elasticity). The tax raises the supply curve to $S + \text{tax}$. The **new higher price P2** is what **consumers** pay; **P3** is what **suppliers** keep. The split of the rectangle between them is the **incidence**:

- **Left — elastic demand ($PED > 1$):** consumers easily cut back, so firms **can't raise the price much** → the **supplier absorbs most** of the tax.
- **Right — inelastic demand ($PED < 1$):** consumers keep buying regardless, so the firm **passes most** of the tax through → the **consumer bears most** of it.
- **Policy take-aways:** (1) "sin taxes" on inelastic goods (cigarettes, alcohol) fall mostly on **consumers** and raise a lot of revenue with **little DWL** (quantity barely moves) — which is *why* governments like them, and why they can be **regressive** (hit the poor). (2) We **prefer taxes on inelastic bases** — they distort choices least.
- **Putting a number on the split — the pass-through formula.** The diagrams say *which* side pays more; this formula says *how much*. Each side's share of the tax is the **other side's** elasticity over the total:

TAX INCIDENCE (PASS-THROUGH)

$$\text{consumer share} = E_S \div (E_S + |E_D|) \quad \cdot \quad \text{producer share} = |E_D| \div (E_S + |E_D|)$$

- **Read it as "the other side's elasticity on top":** the **consumer** share carries **E_S** (supply's elasticity); the **producer** share carries **$|E_D|$** (demand's elasticity). Same denominator; the two shares **add to 1** (they split the whole tax — a handy check). - **Why the "other" side's elasticity?** A side pays a big share when it **can't escape** — i.e. when it's inelastic **and the other side is elastic**. If

supply is very elastic (big E_S), sellers walk away rather than absorb the tax, so it lands on **consumers** $\rightarrow E_S$ on top of the consumer share. This is the same "inelastic side bears more" rule from the diagrams, now as an equation. (Worked in [practice problem 11](#micro3-practice).) - **Subsidies are the mirror image:** $p_s = p_b + s$ — a subsidy lowers the buyer price, raises the seller price, **expands** quantity above the efficient level, and also creates a **deadweight loss** (units produced that cost more than consumers value them). Its incidence, too, depends on elasticities (inelastic demand — e.g. emergency care, primary education — $\rightarrow$ small DWL). - **Worked example** (subsidy $s = €3$): **same market** ($Q_D = 23 - p_b$, $Q_S = -1 + p_s/2$) but now $p_s = p_b + 3$ $\rightarrow$ solving gives $p_b = 15$, $p_s = 18$, $Q = 8$ — buyers pay **less** and sellers get **more**, and quantity **rises** above the no-subsidy level ($6 \rightarrow 8$). The subsidy costs the government $s \times Q = 3 \times 8 = €24$, part of which is **deadweight loss**.

Cases & examples

Read each as a worked use of an elasticity number (or a tax result).

- **Sugary-soda tax (Allcott et al. 2019)**. Own-price elasticity ≈ -1.37 (elastic). A 5% price rise $\rightarrow$ $\sim 7\%$ fall in quantity; to cut demand 15% you'd raise price $\sim 11\%$. $\rightarrow$ *Illustrates: elasticity as the **design tool** for a corrective ("sin") tax.*
- **World cocoa demand**. Inelastic (≈ -0.19 to -0.96 ; policy estimate -0.34). $\rightarrow$ *Illustrates: **inelastic demand** $\rightarrow$ supply shocks hit price, so bad harvests swing incomes for growers in Ghana / Côte d'Ivoire.*
- **US beer — brand vs category (Miller et al. 2021)**. Product-level elasticity ≈ -4.74 , category ≈ -0.60 . $\rightarrow$ *Illustrates: **aggregation** — a single brand has close substitutes (elastic), the category doesn't (inelastic).*
- **Sweden–Denmark cross-border alcohol (Asplund et al. 2007)**. Swedes drive to Denmark for cheaper drink. Estimated for spirits: **own-price** -1.3 (elastic), **cross-price (Danish price)** $+0.3$ (substitute), **income** $+1.4$ (normal good). $\rightarrow$ *Illustrates: **all three elasticities at once** — own, cross, income — from real data.*
- **Electricity demand (Labandeira et al. 2017)**. ≈ -0.24 short-run vs -0.6 long-run. $\rightarrow$ *Illustrates: demand is **more elastic in the long run** (time to adjust equipment/habits).*

Formula sheet — quick reference

Quantity	Formula	Sign / meaning
Price elasticity of demand	$E_D = \% \Delta Q_D \div \% \Delta p$	negative ; $ E > 1$ elastic, $ E < 1$ inelastic
Point elasticity (linear)	$E_D = (dQ/dp) \cdot (p/Q)$	varies along the curve; more elastic at higher p
Constant-elastic demand	$Q = a \cdot p^\epsilon \rightarrow \ln Q = \ln a + \epsilon \ln p$	elasticity = ϵ everywhere
Cross-price elasticity	$E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B$	+ substitutes, – complements
Income elasticity	$E_{\text{income}} = \% \Delta Q \div \% \Delta I$	+ normal, – inferior ; food < 1 (Engel)
Supply elasticity	$E_S = \% \Delta Q_S \div \% \Delta p$	positive
Tax wedge	$p_b = p_s + t$	buyers pay more, sellers get less; DWL
Tax revenue	$t \times Q_{\text{taxed}}$	rectangle; rest of loss = deadweight loss
Subsidy wedge	$p_s = p_b + s$	expands Q above efficient; also a DWL

Exam pointers

- Compute an elasticity from two (p, Q) points and **name** it (elastic/inelastic/unit) — the –1/3 example.
- Use the point-elasticity trick $E_D = (dQ/dp) (p/Q)$ on a **linear demand** and show elasticity **changes along the curve** (more elastic at higher price).
- Give the **sign rules** for cross-price (substitutes/complements) and income (normal/inferior) elasticities, with an example each.
- Solve a **unit-tax market** ($p_b = p_s + t$) for p_b , p_s , Q , compute **revenue** and **deadweight loss** — the t=€3 example.
- State and **explain** the two big tax results: **legal ≠ economic incidence** (doesn't matter who's taxed) and **incidence depends on relative elasticities** (inelastic side pays more); use the sin-tax application.
- Explain **why elasticity is unit-free** and why demand is **more elastic in the long run** and for narrower product categories.

Bridge. Lecture 3 turns the Chapter 2 model from *qualitative* ("the curve shifts left") into *quantitative* ("by how much, and split how between price and quantity"). Elasticity is also the hinge to later chapters: **firm pricing** (a monopolist's markup depends on E_D), **tax policy & deadweight loss** (Ch. 7), and **who gains from trade**.

★ Practice problems — the whole chapter (with worked solutions)

A set covering **every calculation** in this lecture. Try each closed-book first, then check against the worked solution. (Exam is open-book, no AI — so the skill being tested is doing these fast and correctly, not memorising.)

What you must be able to calculate (each maps to a problem below):

Skill	Formula	Problem
Own-price elasticity from two points + name it	$E_D = \% \Delta Q \div \% \Delta p$	1
Predict $\% \Delta Q$ from $\% \Delta p$ (and reverse)	$\% \Delta Q = E_D \cdot \% \Delta p$	2
Point elasticity on a linear curve (+ find the unit-elastic point)	$E_D = (dQ/dp) \cdot (p/Q)$	3
Elasticity → total revenue (raise or cut price?)	$TR = p \cdot Q$	4
Constant-elastic demand	$Q = a \cdot p^\epsilon$	5
Cross-price elasticity + classify	$E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B$	6
Income elasticity + classify	$E_{\text{income}} = \% \Delta Q \div \% \Delta I$	7
Supply elasticity	$E_S = \% \Delta Q_S \div \% \Delta p$	8
Unit tax: solve p_b , p_s , Q , revenue, DWL, incidence	$p_b = p_s + t$	9
Subsidy: solve p_b , p_s , Q , cost, DWL	$p_s = p_b + s$	10
Incidence from elasticities (pass-through)	$\text{consumer share} = E_S \div (E_S + E_D)$ · $\text{producer share} = E_D \div (E_S + E_D)$	11
Equilibrium + a price control (Ch. 2 recap)	set $Q_D = Q_S$	12

Problems

1. A bakery raises bread from €4 → €5; weekly sales fall from 100 → 80 loaves. Find the price elasticity of demand and say whether demand is elastic, inelastic, or unit-elastic.
2. (a) The own-price elasticity of petrol is -0.3 . If the price rises 10%, by how much does quantity demanded change? (b) A cinema wants ticket sales to rise 9%; demand elasticity is -1.5 . What price change is needed?

3. Demand is $Q = 120 - 4p$. (a) Find E_D at $p = 10$. (b) Find E_D at $p = 20$. (c) At what price is demand **unit-elastic**?
4. Using the same demand $Q = 120 - 4p$: at $p = 10$ and at $p = 20$, should the firm **raise or cut** price to increase **total revenue**? At what price is revenue **maximised**?
5. Demand is $Q = 500 \cdot p^{-2}$. (a) What is the elasticity of demand? (b) Find quantity at $p = 5$ and $p = 10$. (c) If price rises 1% from €5, roughly what is $\% \Delta Q$?
6. Classify each pair as substitutes, complements, or unrelated, and give E_{cross} : (a) price of **tea** +8% → demand for **coffee** +4%; (b) price of **printers** +10% → demand for **ink** +6% **fall**; (c) price of **umbrellas** +10% → demand for **coffee** 0%.
7. Classify the good and give E_{income} : (a) income +5% → **restaurant meals** +15%; (b) income +10% → **bus tickets** -4%; (c) income +8% → **bread** +2%.
8. When the price rises from €8 → €10, quantity supplied rises from 60 → 69. Find the price elasticity of supply and classify it.
9. A market has $Q_D = 100 - 4p$ and $Q_S = -20 + 2p$. (a) Find the pre-tax equilibrium. (b) A **unit tax** $t = €6$ is imposed. Solve for p_b , p_s , and Q . (c) Find **tax revenue** and the **deadweight loss**. (d) How is the burden split between buyers and sellers, and **why**?
10. Same market ($Q_D = 100 - 4p$, $Q_S = -20 + 2p$). Now the government pays a **subsidy** $s = €6$ per unit. Solve for p_b , p_s , Q , the **cost to government**, and the **deadweight loss**.
11. At the pre-tax equilibrium of problem 9, the elasticities are $E_D = -4$ and $E_S = +2$. Use the pass-through formula to find the **share of the tax borne by consumers vs producers**, and check it against your answer to 9(d).
12. (Ch. 2 recap.) $Q_D = 80 - 2p$, $Q_S = -10 + p$. (a) Find equilibrium p^* and q^* . (b) The government sets a **price ceiling** at €25. Compute the resulting **shortage**.

Solutions

1. $\% \Delta Q = (80 - 100)/100 = -20\%$; $\% \Delta p = (5 - 4)/4 = +25\%$.

$$E_D = -20\% \div +25\% = -0.8$$

$|-0.8| < 1 \rightarrow$ ****inelastic**** (quantity moves less than price). * (Note: revenue actually rises — see problem 4's logic.)*

2. (a) $\% \Delta Q = E_D \cdot \% \Delta p = -0.3 \times 10\% = -3\%$ (quantity falls 3%). (b) rearrange: $\% \Delta p = \% \Delta Q \div E_D = 9\% \div (-1.5) = -6\% \rightarrow$ the cinema must **cut** price 6%.
3. Slope $dQ/dp = -4$ (constant). Use $E_D = (dQ/dp) \cdot (p/Q)$.
- (a) $p = 10 \rightarrow Q = 120 - 40 = 80 \rightarrow E_D = -4 \times 10/80 = -0.5$ (inelastic).
 - (b) $p = 20 \rightarrow Q = 120 - 80 = 40 \rightarrow E_D = -4 \times 20/40 = -2$ (elastic).
 - (c) unit-elastic: set $-4 \cdot (p/Q) = -1 \rightarrow Q = 4p \rightarrow 120 - 4p = 4p \rightarrow 8p = 120 \rightarrow p = 15$ ($Q = 60$). (On any straight-line demand, the unit-elastic point is the **midpoint**.)

4. Revenue $TR = p \cdot Q$.

- At $p = 10$ (inelastic, $|E| < 1$): raising price **raises** revenue. Check: $p=10 \rightarrow TR = 10 \times 80 = 800$; $p=11 \rightarrow Q = 76$, $TR = 836$. $\rightarrow$ **raise price**.
- At $p = 20$ (elastic, $|E| > 1$): cutting price **raises** revenue. Check: $p=20 \rightarrow TR = 800$; $p=19 \rightarrow Q = 44$, $TR = 836$. $\rightarrow$ **cut price**.
- Revenue is maximised at the unit-elastic point** $p = 15$: $TR = 15 \times 60 = 900$. (Rule: $|E| < 1 \rightarrow$ raise price; $|E| > 1 \rightarrow$ cut price; max TR where $|E| = 1$.)

5. (a) For $Q = a \cdot p^\epsilon$, the elasticity is the exponent: $E_D = -2$ (constant at every price). (b) $p = 5 \rightarrow Q = 500/25 = 20$; $p = 10 \rightarrow Q = 500/100 = 5$. (c) $\% \Delta Q \approx E_D \cdot \% \Delta p = -2 \times 1\% = -2\%$.

6. $E_{cross} = \% \Delta Q_A \div \% \Delta p_B$. (a) $+4 \div +8 = +0.5 \rightarrow$ **substitutes** (tea dearer $\rightarrow$ more coffee). (b) $-6 \div +10 = -0.6 \rightarrow$ **complements** (printers dearer $\rightarrow$ less ink). (c) $0 \div +10 = 0 \rightarrow$ **unrelated**.

7. $E_{income} = \% \Delta Q \div \% \Delta I$. (a) $+15 \div +5 = +3 \rightarrow$ **normal, and a luxury** ($E > 1$). (b) $-4 \div +10 = -0.4 \rightarrow$ **inferior**. (c) $+2 \div +8 = +0.25 \rightarrow$ **normal necessity** ($0 < E < 1$; cf. Engel's law for food).

8. $\% \Delta Q_S = (69 - 60)/60 = +15\%$; $\% \Delta p = (10 - 8)/8 = +25\%$.

$$E_S = +15\% \div +25\% = +0.6$$

$0 < E_S < 1 \rightarrow$ **inelastic supply** (output hard to expand quickly).

9. (a) **Pre-tax**: set $Q_D = Q_S$: $100 - 4p = -20 + 2p \rightarrow 120 = 6p \rightarrow p^* = 20, Q^* = 20$.

- (b) **With** $t = 6$, use $p_b = p_s + 6$, $Q_D = 100 - 4p_b$, $Q_S = -20 + 2p_s$. Set equal: $100 - 4(p_s + 6) = -20 + 2p_s \rightarrow 76 - 4p_s = -20 + 2p_s \rightarrow 96 = 6p_s \rightarrow p_s = 16, p_b = 22, Q = 100 - 4(22) = 12$. (check $Q_S = -20 + 32 = 12 \checkmark$)
- (c) **Tax revenue** = $t \times Q = 6 \times 12 = \text{€}72$. **DWL** = $\frac{1}{2} \times t \times (Q^* - Q_{tax}) = \frac{1}{2} \times 6 \times (20 - 12) = \text{€}24$.
- (d) Buyers bear $p_b - p^* = 22 - 20 = \text{€}2$; sellers bear $p^* - p_s = 20 - 16 = \text{€}4$ (they sum to $t = 6$). **Sellers bear more because supply is the more inelastic side** (slope 2 vs demand's 4 $\rightarrow$ demand is more elastic, so it escapes more of the tax onto quantity).

10. **With** $s = 6$, use $p_s = p_b + 6$: $100 - 4p_b = -20 + 2(p_b + 6) \rightarrow 100 - 4p_b = -8 + 2p_b \rightarrow 108 = 6p_b \rightarrow p_b = 18, p_s = 24, Q = 100 - 4(18) = 28$. (check $Q_S = -20 + 48 = 28 \checkmark$) — buyers pay less ($18 < 20$), sellers get more ($24 > 20$), and **Q rises** ($20 \rightarrow 28$). **Cost to government** = $s \times Q = 6 \times 28 = \text{€}168$. **DWL** = $\frac{1}{2} \times s \times (Q_{sub} - Q^*) = \frac{1}{2} \times 6 \times (28 - 20) = \text{€}24$ (over-production: units that cost more than they're worth).

11. Pass-through shares:

INCIDENCE FROM ELASTICITIES

$$\text{consumer share} = E_S \div (E_S + |E_D|) \quad \cdot \quad \text{producer share} = |E_D| \div (E_S + |E_D|)$$

With $E_S = 2$ and $|E_D| = 4$: **consumer share** = $2 \div (2+4) = 1/3$, **producer share** = $4 \div 6 = 2/3$. Of the €6 tax that's **€2 on buyers, €4 on sellers** — exactly matching 9(d). $\checkmark$ (Rule: the **more inelastic** side pays the bigger share.)

12. (a) $80 - 2p = -10 + p \rightarrow 90 = 3p \rightarrow p^* = 30, q^* = 20$. (b) At the ceiling $p = 25$: $Q_D = 80 - 50 = 30$, $Q_S = -10 + 25 = 15 \rightarrow \text{shortage} = Q_D - Q_S = 30 - 15 = 15 \text{ units}$.

★ Fun facts & memorable details (Lecture 3)

Sticky bits from Elasticities & Taxes.

- **It doesn't matter who you tax.** In a competitive market the burden of a tax is **identical** whether the government bills the shop or the shopper — "all that matters is that the tax drives a wedge." (This is why arguing about *who* a tax is "on" often misses the point.)
 - **Why cigarette & alcohol taxes raise so much money:** demand is **inelastic**, so people keep buying — the tax is **passed to consumers** with little drop in quantity (and is often **regressive**).
 - **The soda-tax number:** the own-price elasticity of sugary drinks (≈ -1.37) is a real input to real policy — a 5% price rise cuts consumption $\sim 7\%$.
 - **Engel's law (1857):** the **richer** you get, the **smaller** the share of income you spend on **food** — one of the oldest empirical regularities in economics, and still used to gauge living standards.
 - **Pharma = "defensive stock":** demand for medicine barely moves with income (income-inelastic), so drug companies hold up in recessions.
 - **Elasticity has a time dimension:** when petrol jumps, you can't do much *this week* (inelastic), but over years you buy a smaller car and move closer to work (elastic) — the "short-run vs long-run" split.
 - **Brand vs category:** demand for *a* beer is wildly elastic (≈ -4.7 — switch brands!), but demand for *beer* is inelastic (≈ -0.6) — same drink, opposite elasticity, just different zoom level.
 - **Swedes really do drive to Denmark for cheap booze** — enough that economists measured the cross-border **cross-price elasticity** of spirits ($+0.3$) from the data (Asplund et al. 2007).
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